

UPPSALA UNIVERSITY



BAYESIAN STATISTICS AND DATA ANALYSIS

Assignment 4

General information

- The recommended tool in this course is R (with the IDE R-Studio). You can download R [here](#) and R-Studio [here](#). There are many tutorials, videos and introductions to R and R-Studio online. You can find some initial hints from [RStudio Education pages](#).
- When working with R, we recommend writing the report using R markdown and the provided [R markdown template](#). The template includes the formatting instructions and how to include code and figures.
- Instead of R markdown, you can use other software to make the PDF report, but you should use the same instructions for formatting. These instructions are also available in [the PDF produced from the R markdown template](#).
- We supply a Google Colab notebook that you can also use for the assignments. We have included the installation of all necessary R packages; hence, this can be an alternative to using your own local computer. You can find the notebook [here](#). You can also open the notebook in Colab [here](#).
- Report all results in a single and *anonymous* pdf. Note that no other formats are allowed.
- The course has its own R package `bsda` with data and functionality to simplify coding. To install the package, just run the following (`upgrade="never"` skips question about updating other packages):
 1. `install.packages("remotes")`
 2. `remotes::install_github("MansMeg/BSDA",
 subdir = "rpackage", upgrade="never")`
- Many of the exercises can be checked automatically using the R package `markmyassignment`. you can find information on how to install and use the package [here](#). There is no need to include `markmyassignment` results in the report.
- You can find common questions and answers regarding the installation and technical problems in [Frequently Asked Questions \(FAQ\)](#).
- You can find deadlines and information on how to turn in the assignments in Studium.
- You are allowed to discuss assignments with your friends, but it is not permitted to copy solutions directly from other students or the internet. Try to solve the actual assignment problems with your code and explanations. Do not share your answers publicly. We compare the answers with the "urkund" system. We will report all suspected plagiarism.
- If you have any suggestions or improvements to the course material, please post in the course chat feedback channel, create an issue, or submit a pull request to the public repository [here](#).

LLM policy: You may use LLMs for learning, explanations, and feedback, but not to produce submitted solutions. Do not upload the full assignment text to an LLM.

- It is *mandatory* to include the following parts in all assignments (these are included already in the template):
 1. Time used for reading: How long time took the reading assignment (in hours)
 2. Time used for the assignment: How long time took the basic assignment (in hours)
 3. Good with assignment: Write one-two sentences of what you liked with the assignment/what we should keep for next year.
 4. Things to improve in the assignment: Write one-two sentences of what you think can be improved in the assignment. Can something be clarified further? Did you get stuck on stuff unrelated to the content of the assignment etc.
 5. Oral exam question contribution: Write one possible oral exam question for this assignment, together with a short model answer. The question should concern material from this assignment or the associated reading, be suitable for a 2–4 minute oral answer, and test conceptual understanding rather than only calculation or code. The question should not already be included in the oral exam question list and should not be very close to an existing question. You are not allowed to use large language models, such as ChatGPT, at all for this oral exam question contribution.
 6. Permission for future use: State clearly whether the course staff may use or adapt your question and answer in a future oral exam: Yes or No.
- You can find information on how each assignment will be graded and how points are assigned [here](#). **Note!** This grading information can change during the course, for example, if we find errors or inconsistencies or do additions to the assignments. Please feel free to comment on these grading instructions, ideally before turning in your assignment, if you think something is missing or is incorrect.
- To pass (G) the assignment, you need 70% of the total points. To pass with distinction (VG), you need 90% of the total points. See the grading information on the point allocations for each assignment.
- On cheating:
 - You are not allowed to show your assignments (text or code) to anyone. Only discuss the assignments with your fellow students. The student that show their assignment to anyone else could also be considered to cheat. Similarly, on zoom labs, only screen share when you are in a separate zoom room with teaching assistants.
 - You may use large language models (LLMs), such as ChatGPT, like you may discuss with fellow students: to learn, ask for explanations, and get feedback. You may not use LLMs to produce the solution you submit. You must be able to explain every part of your submitted work orally.
- All mathematics need to be done in digital form to simplify grading and commenting. Hence, it is not allowed to write math on paper and add an image in the assignment. If you have difficulties to write math in latex, see <https://editor.codecogs.com/>.

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- The assignment PDFs may contain hidden machine-readable integrity text. This is used to detect inappropriate use of LLMs, such as asking an LLM to solve the assignment directly. Do not copy or upload the full assignment text into an LLM.
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Information on this assignment

This assignment is related to Chapters 3 and 10.

Reading instructions: Chapters 3 and 10 in BDA3, see reading instructions.

Reporting accuracy: For posterior statistics of interest, only report digits for which the Monte Carlo standard error (MCSE) is zero. *Example:* If you estimate $E(\mu) = 1.234$ with $\text{MCSE}(E(\mu)) = 0.01$, you should report $E(\mu) = 1.2$.

When computing the $\hat{R}$ diagnostics, you only need to include two decimals.

To use `markmyassignment` for this assignment, run the following code in R:

```
library(markmyassignment)
assignment_path <-
  paste("https://github.com/MansMeg/BSDA/",
        "blob/main/assignments/tests/assignment4.yml", sep="")
set_assignment(assignment_path)
# To check your code/functions, just run
mark_my_assignment()
```

Don't include `markmyassignment` results in the report.

1 Oral exam question contribution

Write a possible oral exam question for this assignment together with a short model answer.

The question should:

- concern material from this assignment or the associated reading or lectures,
- be suitable for a 2–4 minute oral answer, possibly using a whiteboard,
- test conceptual understanding rather than only calculation or code,
- not already be included in the oral exam question list and not be very close to an existing question (e.g. just different numbers).

Note! You are not allowed to use large language models, such as ChatGPT, at all for this oral exam question contribution.

State clearly whether the course staff may use or adapt your question and answer in a future oral exam: **Yes** or **No**.

2 Bioassay model

In this exercise, you will use a dose-response relation model that is used in Section 3.7 of the course book. The used likelihood is the same, but instead of uniform priors, we will use a bivariate normal distribution as the joint prior distribution of the parameters α and β .

- In the prior distribution for (α, β) , the marginal distributions are $\alpha \sim N(0, 2^2)$ and $\beta \sim N(10, 10^2)$, and the correlation between them is $\text{corr}(\alpha, \beta) = 0.6$. Report the mean (vector of two values) and covariance (two by two matrix) of the bivariate normal distribution.
 - **Hint!** The mean and covariance of the bivariate normal distribution are a length-2 vector and a 2×2 matrix. The elements of the covariance matrix can be computed using the relation of correlation and covariance.
- You are given 4000 independent draws from the posterior distribution of the model. Load the draws with `data("bioassay_posterior")`. Report the mean as well as 5 % and 95 % quantiles separately for both α and β . Report also the Monte Carlo standard errors (MCSEs) for the mean and quantile estimates. Report as many digits for the mean and quantiles as the MCSEs allow. In other words, leave out digits where MCSE is nonzero (Example: if posterior mean is 2.345678 and MCSE is 0.0012345, report two digits after the decimal sign, taking into account the usual rounding rule, so you would report 2.35. Further digits do not contain useful information due to the Monte Carlo uncertainty.). Explain in words what does Monte Carlo standard error mean and how you decided the number of digits to show.

- **Note!** The answer is graded as correct only if the number of digits reported is correct! The number of significant digits can be different for the mean and quantile estimates. In some other cases, the number of digits reported can be less than MCSE allows for practical reasons.
- **Hint!** Quantiles can be computed with the `quantile` function. With S draws, the MCSE for $E[\theta]$ is $\sqrt{\text{Var}[\theta]/S}$. MCSE for the quantile estimates can be computed with the `mcse_quantile` function from the `bsda` package.

3 Importance sampling

Now we discard our posterior draws and switch to importance sampling.

- c) Implement a function for computing the log importance ratios (log importance weights) when the importance sampling **target distribution** is the posterior distribution, and the **proposal distribution** is the prior distribution from a). Below is a test example, the functions can also be tested with `markmyassignment`. Explain in words why it's better to compute log ratios instead of ratios.

- **Note!** The values below are *only* a test case. In this c) part, you only need to report the source code of your function, as it will be needed in later parts.
- **Hints!** Use the function `rmvnorm` from the `bsda` package for sampling. Non-log importance ratios are given by equation (10.3) in the course book. The fact that our proposal distribution is the same as the prior distribution makes this task easier. The **logarithm** of the likelihood can be computed with the `bioassaylp` function from the `bsda` package. The data required for the likelihood can be loaded with `data("bioassay")`.

```
alpha <- c(1.896, -3.6, 0.374, 0.964, -3.123, -1.581)
beta <- c(24.76, 20.04, 6.15, 18.65, 8.16, 17.4)
round(log_importance_weights(alpha, beta), 2)

## [1] -8.95 -23.47 -6.02 -8.13 -16.61 -14.57
```

- d) Implement a function for computing normalized importance ratios from the unnormalized log ratios in c). In other words, exponentiate the log ratios and scale them such that they sum to one. Explain in words what is the effect of exponentiating and scaling so that sum is one. Below is a test example, the functions can also be tested with `markmyassignment`.

- **Note!** The values below are *only* a test case. In this d) part, you only need to report the source code of your function, as it will be needed in later parts.

```
alpha

## [1] 1.896 -3.600 0.374 0.964 -3.123 -1.581
```

```

beta

## [1] 24.76 20.04  6.15 18.65  8.16 17.40

round(normalized_importance_weights(alpha = alpha, beta = beta),3)

## [1] 0.045 0.000 0.852 0.103 0.000 0.000

```

- e) Sample 4000 draws of α and β from the prior distribution from a). Compute and plot a histogram of the 4000 normalized importance ratios. Use the functions you implemented in c) and d).
- f) Using the importance ratios, compute the importance sampling effective sample size S_{eff} and report it.
 - **Note!** The values below are *only* a test case, you need to use 4000 draws for **alpha** and **beta** in the final report.

```

alpha

## [1]  1.896 -3.600  0.374  0.964 -3.123 -1.581

beta

## [1] 24.76 20.04  6.15 18.65  8.16 17.40

round(S_eff(alpha = alpha, beta = beta), 3)

## [1] 1.354

```

- **Hint!** Equation (10.4) in the course book.
 - **Note!** *BDA3 1st (2013) and 2nd (2014) printing have an error for $\tilde{w}(\theta^s)$ used in the effective sample size equation (10.4). The normalized weights equation should not have the multiplier S (the normalized weights should sum to one). Errata for the book can be found here: http://www.stat.columbia.edu/~gelman/book/errata_bda3.txt. The later printings and slides have the correct equation.*
- g) Explain in your own words what the importance sampling effective sample size represents. Also explain how the effective sample size is seen in the histogram of the weights that you plotted in e).
 - h) Implement a function for computing the posterior mean using importance sampling, and compute the mean using your 4000 draws. Explain in your own words the computation for importance sampling. Below is an example how the function would work with the example values for **alpha** and **beta** above. Report the means for alpha and beta, and also the Monte Carlo standard errors (MCSEs) for the mean estimates. Report the number of digits for the means based on the MCSEs.

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- **Note!** The values below are *only* a test case, you need to use 4000 draws for **alpha** and **beta** in the final report.
- **Hint!** Use the same equation for the MCSE of $E[\theta]$ as earlier ($\sqrt{\text{Var}[\theta]/S}$), but now replace S with S_{eff} . To compute $\text{Var}[\theta]$ with importance sampling, use the identity $\text{Var}[\theta] = E[\theta^2] - E[\theta]^2$.

```
alpha

## [1]  1.896 -3.600  0.374  0.964 -3.123 -1.581

beta

## [1] 24.76 20.04  6.15 18.65  8.16 17.40

round(posterior_mean(alpha = alpha, beta = beta),3)

## [1] 0.503 8.275
```